

```
mysql> CREATE DATABASE college_detail;  
Query OK, 1 row affected (0.04 sec)
```

```
mysql> USE college_detail;  
Database changed
```

```
mysql> CREATE TABLE college (  
->     serialno INT PRIMARY KEY,  
->     name VARCHAR(30),  
->     department VARCHAR(30),  
->     year INT,  
->     marks INT  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> INSERT INTO college VALUES (1, 'Ram', 'CSE', 2, 85);  
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO college VALUES (2, 'Shyam', 'ECE', 3, 78);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO college VALUES (3, 'John', 'EEE', 2, 90);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO college VALUES (4, 'Priya', 'CSE', 1, 88);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO college VALUES (5, 'Kavi', 'ECE', 3, 76);  
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT * FROM college;
```

serialno	name	department	year	marks
1	Ram	CSE	2	85
2	Shyam	ECE	3	78
3	John	EEE	2	90
4	Priya	CSE	1	88
5	Kavi	ECE	3	76

```
mysql> ALTER TABLE college
-> ADD mark1 INT,
-> ADD mark2 INT,
-> ADD mark3 INT,
-> ADD average_marks DECIMAL(5,2);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> UPDATE college
-> SET average_marks = (mark1 + mark2 + mark3) / 3;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 5 Changed: 0 Warnings: 0
```

```
mysql> SELECT name, average_marks
-> FROM college
-> WHERE average_marks > 70;
Empty set (0.00 sec)
```

```
mysql> SELECT * FROM college;
```

serialno	name	department	year	marks	mark1	mark2	mark3	average_marks
1	Ram	CSE	2	85	NULL	NULL	NULL	NULL
2	Shyam	ECE	3	78	NULL	NULL	NULL	NULL
3	John	EEE	2	90	NULL	NULL	NULL	NULL
4	Priya	CSE	1	88	NULL	NULL	NULL	NULL
5	Kavi	ECE	3	76	NULL	NULL	NULL	NULL

```
5 rows in set (0.00 sec)
```

```
mysql> UPDATE college
-> SET mark1 = 80, mark2 = 85, mark3 = 90
-> WHERE name = 'Ram';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> UPDATE college
      -> SET mark1 = 80, mark2 = 85, mark3 = 90
      -> WHERE name = 'Ram';
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE college
      -> SET mark1 = 70, mark2 = 75, mark3 = 72
      -> WHERE name = 'Shyam';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE college
      -> SET mark1 = 88, mark2 = 92, mark3 = 86
      -> WHERE name = 'John';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE college
      -> SET mark1 = 65, mark2 = 68, mark3 = 70
      -> WHERE name = 'Priya';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> UPDATE college
      -> SET mark1 = 78, mark2 = 74, mark3 = 80
      -> WHERE name = 'Kavi';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```



```
mysql> select name and marks,AVG(marks) AS avg_marks
-> FROM student
-> ;
ERROR 1146 (42S02): Table 'college_detail.student' doesn't exist
```

```
mysql> SELECT name, marks
-> FROM college
-> WHERE marks > (
->     SELECT AVG(marks)
->     FROM college
-> );
```

name	marks
Ram	85
John	90
Priya	88

```
3 rows in set (0.03 sec)
```

```
mysql>
mysql> UPDATE college
    -> SET mark1 = 78, mark2 = 74, mark3 = 80
    -> WHERE name = 'Kavi';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> UPDATE college
    -> SET average_marks = (mark1 + mark2 + mark3) / 3;
Query OK, 5 rows affected, 4 warnings (0.01 sec)
Rows matched: 5  Changed: 5  Warnings: 4
```

```
mysql> SELECT name, average_marks
    -> FROM college
    -> WHERE average_marks > 70;
```

name	average_marks
Ram	85.00
Shyam	72.33
John	88.67
Kavi	77.33

```
4 rows in set (0.00 sec)
```



```
mysql> SELECT name, average_marks  
-> FROM college  
-> WHERE average_marks = 85;
```

name	average_marks
Ram	85.00

1 row in set (0.00 sec)

```
mysql> SELECT name, average_marks  
-> FROM college  
-> WHERE average_marks > 85;
```

name	average_marks
John	88.67

1 row in set (0.00 sec)

```
mysql>
```

```
mysql> SELECT * FROM college WHERE department = ( SELECT department FROM college WHERE name ='Ram');
```

serialno	name	department	year	marks
1	Ram	CSE	2	85
4	Priya	CSE	1	88

2 rows in set (0.00 sec)

```
mysql> SELECT name, marks  
-> FROM college  
-> WHERE marks = (  
->     SELECT MAX(marks)  
->     FROM college  
-> );
```

name	marks
John	90

1 row in set (0.00 sec)

```
mysql> SELECT * FROM college WHERE department=(SELECT FROM department WHERE AVG(marks)>70);
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version